CampusSaathi AI - Sample College Notes

Department of Computer Science & Engineering - Semester 4

UNIT 1: DATA STRUCTURES AND ALGORITHM ANALYSIS

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1. Asymptotic Notations:
   - Big-O (O): Represents the upper bound of algorithm runtime (Worst Case).
   - Omega: Represents the lower bound of runtime (Best Case).
   - Theta: Represents the tight bound where upper and lower bounds meet.

2. Binary Search Trees (BST):
   - A binary tree where the left subtree contains keys smaller than the node.
   - The right subtree contains keys strictly greater than the root node.
   - Search, insertion, and deletion complexity: O(h) where h is tree height.
   - In a balanced BST (AVL or Red-Black), h = O(log n).
   - In a degenerate BST, worst-case complexity degrades to O(n).

3. Common College Exam Questions:
   - Question: Differentiate between Stack (LIFO) and Queue (FIFO).
   - Question: Explain In-order, Pre-order, and Post-order tree traversals.

UNIT 2: DATABASE MANAGEMENT SYSTEMS (DBMS)

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1. ACID Properties in Transaction Processing:
   - Atomicity: All operations succeed or none are applied (All-or-Nothing).
   - Consistency: Database moves from one valid state to another valid state.
   - Isolation: Concurrent transactions do not interfere with each other.
   - Durability: Committed changes survive power loss or system crash.

2. Database Normalization:
   - 1NF: Eliminates repeating groups; ensures atomic/indivisible values.
   - 2NF: In 1NF and removes partial functional dependencies on candidate keys.
   - 3NF: In 2NF and removes transitive functional dependencies (X -> Y -> Z).
   - BCNF: Stricter version of 3NF; for every X -> Y, X must be a super key.

3. Important Exam Formula:
   - Candidate Keys formula: Minimal super key with uniqueness property.